

Smart Shoe Walking Session Analysis

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File: walking_251119_155426.csv

Duration: 182.3 s

Summary:

- User walked casually first, then increased pace later in the session.
- Cadence was estimated directly from pitch oscillations (zero-crossings).

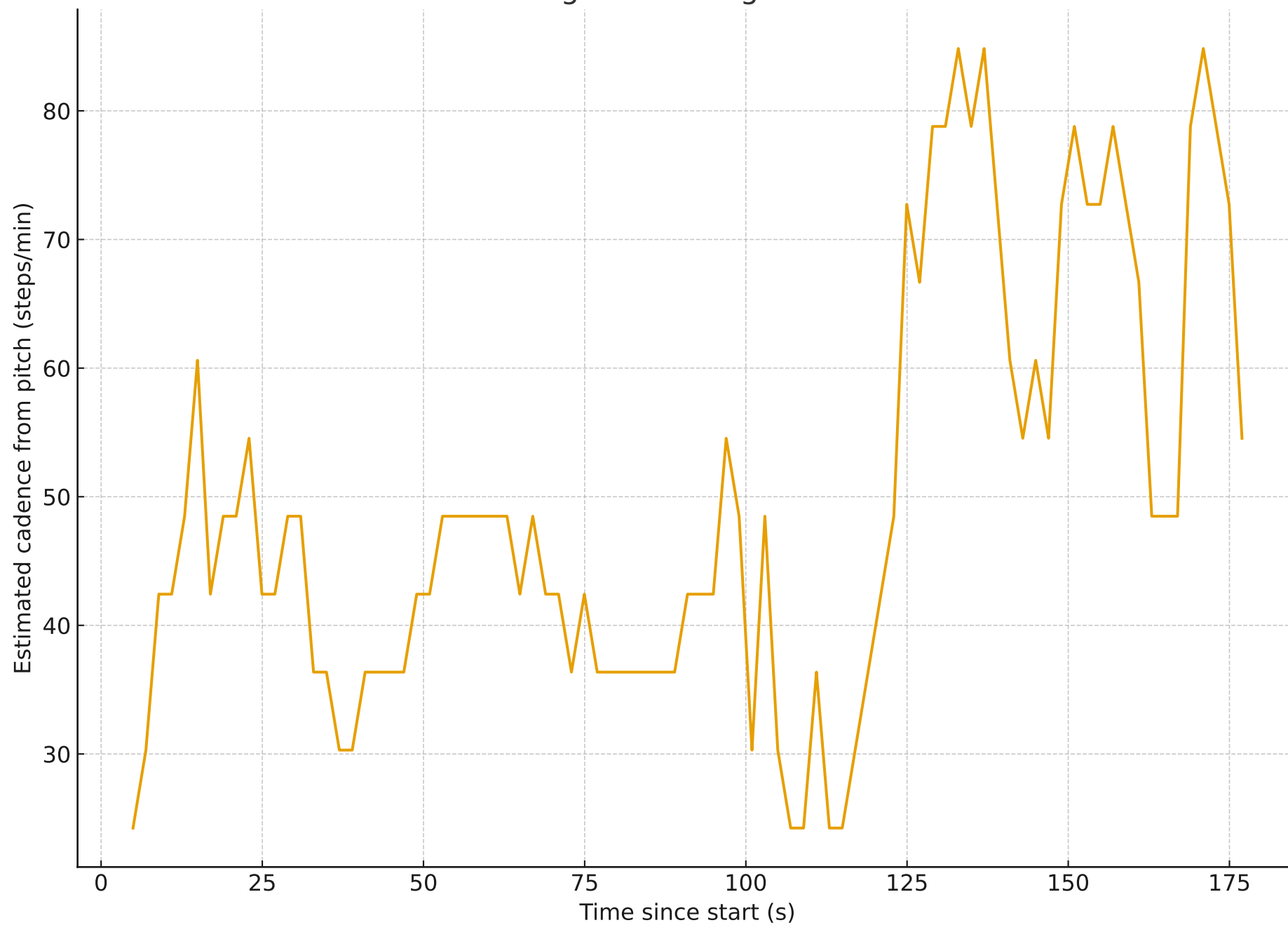
Estimated cadence (20–60 s, casual): 37.6 steps/min

Estimated cadence (120–160 s, faster): 66.2 steps/min

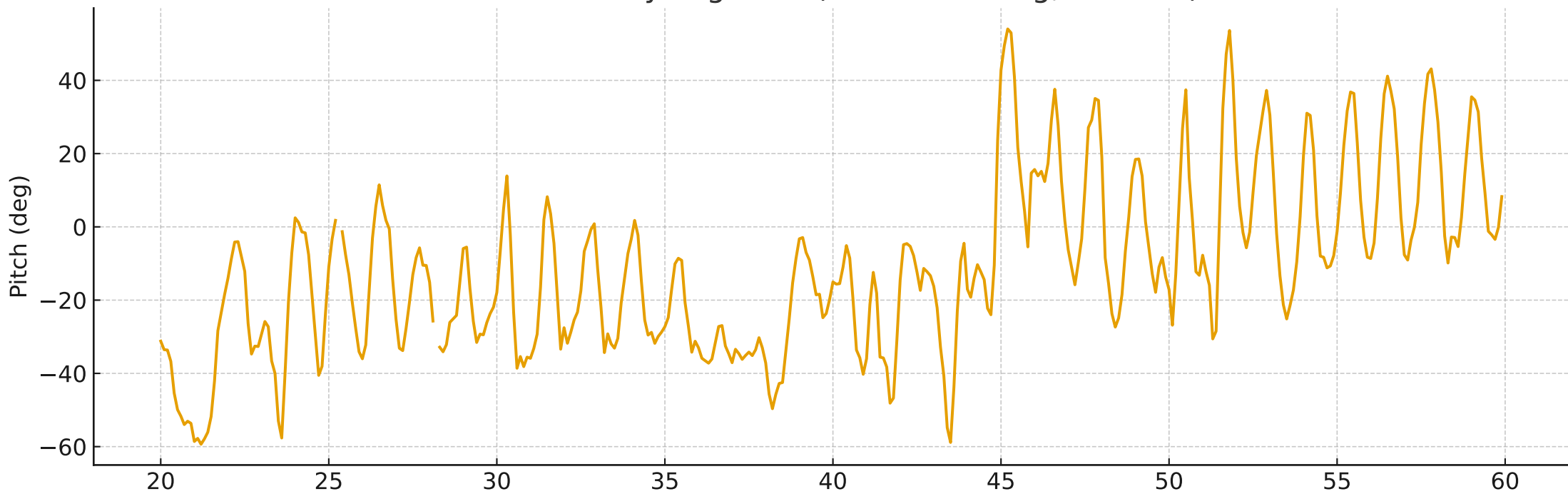
Interpretation:

- Early window shows lower step rate (casual walk).
- Later window shows a clear increase in cadence (~75% higher), consistent with transitioning to a faster walk.

Estimated change in walking cadence over time



Pitch — early segment (casual walking, 20–60 s)



Pitch — later segment (faster walking, 120–160 s)

